

Technical Comparative Analysis:

Landmark-Based Pose Estimation (MediaPipe) vs. End-to-End Convolutional Neural Networks for ASL Recognition

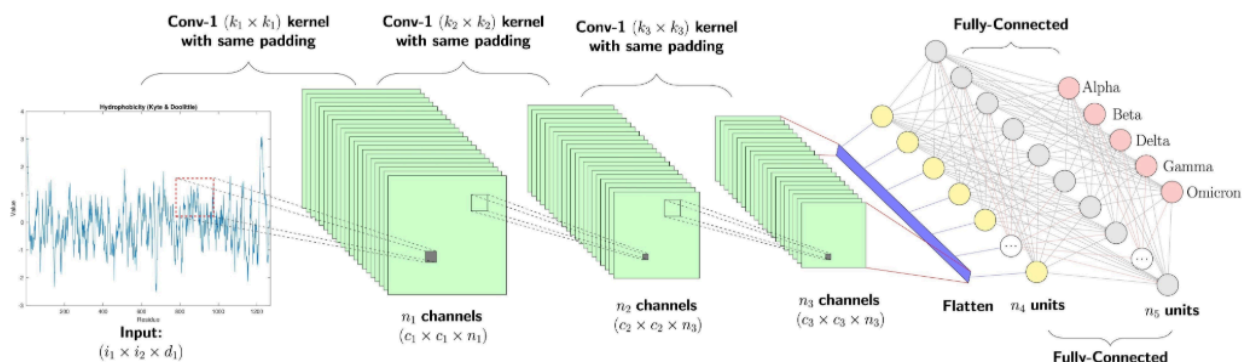
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Project Type: Deep Learning & Computer Vision Research

1. Abstract

In my pursuit of developing robust Digit and ASL Recognition models, I engineered and evaluated two distinct deep learning architectures. This report details my comparative analysis between a Geometric Landmark Model (utilizing MediaPipe) and an End-to-End Convolutional Neural Network (CNN). My research focused on the critical trade-offs between computational latency, environmental robustness, and classification accuracy when deployed in real-time edge-computing scenarios. Through this process, I discovered that while MediaPipe offers "skeletal elegance," the CNN possesses a deeper "visual intuition" that is vital for complex gesture recognition.



multi-dimensional grid of numbers / Tensor in CNN

2. Introduction

The core challenge I addressed in automating ASL Recognition (SLR) is the high degree of spatial variability and the requirement for zero-latency inference. I intentionally developed **two "brains"** for this project to understand different input formats:

Raw pixels vs. Extracted coordinates using 2 different datasets.

My goal was to determine which approach truly bridges the gap between human gestures and machine understanding.

3. Methodology and Architecture

3.1 The Landmark Approach (MediaPipe + DNN)

For the first methodology, I utilized the **MediaPipe Holistic pipeline**. Instead of feeding raw images into my neural network, I used a pre-trained palm detector to identify the hand and a landmark model to map 21 specific 3D points (x, y, z coordinates).

- **Data Preparation:** I flattened these 21 landmarks into a 63-dimensional feature vector per frame.
- **Neural Network Brain:** I built a Deep Neural Network (DNN) using Dense layers and **Dropout** (0.2) to prevent overfitting.
- **Input Shape:** (None, 63).

3.2 The CNN Approach (Raw Pixel Processing)

The second methodology involved a traditional **Convolutional Neural Network (CNN)**. Here, I allowed the model to perform its own feature extraction directly from the raw pixels.

- **Architecture:** I designed a multi-layer stack consisting of **Conv2D** filters (32 and 64), **Batch Normalization** for learning stability, and **MaxPooling2D** for spatial downsampling.
- **Data Augmentation:** To make the CNN more robust, I implemented an **ImageDataGenerator** to perform rotations, zooms, and horizontal flips.
- **Input Shape:** (28, 28, 1).

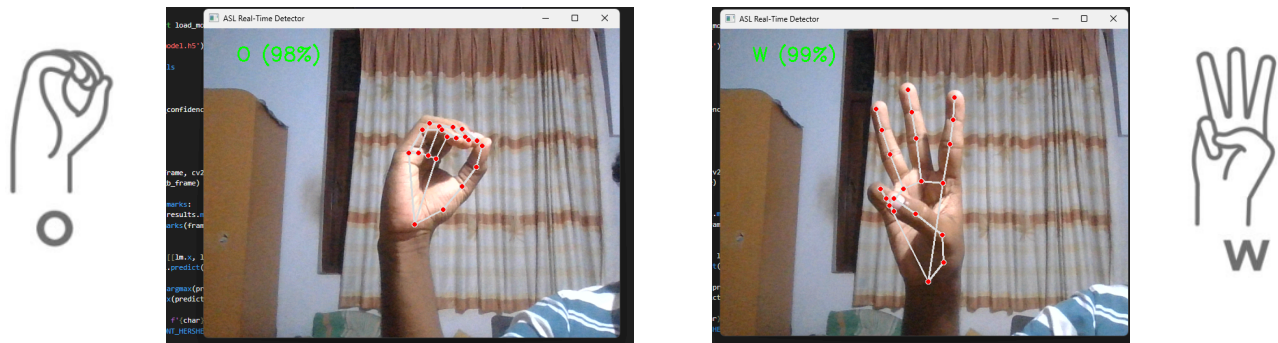
4. Comparative Analytics: What MediaPipe Misses

During my research, I identified 3 critical areas where MediaPipe's abstraction loses vital information that the CNN successfully captures:

1. Information Density and Resolution Thresholds: MediaPipe reduces the hand to a simplified mathematical blueprint, while a CNN retains the full "information density" of the raw pixels.

- **What's missing:** MediaPipe reduces the hand to just 63 coordinates and requires high-resolution input (above 128x128) to maintain tracking stability. A CNN retains, thousands of data points and can successfully extract features even from low-resolution 28x28 images.
- **The Impact:** While MediaPipe is a "spatial filter" that is excellent for high-speed tracking(Real-Time Tracking), it lacks the CNN's "**long-term feature extraction potential**". The CNN's ability to learn from raw pixel data allows it to identify a sign

based on its unique silhouette and internal gradients, providing better answers in hardware-constrained or low-resolution scenarios where MediaPipe fails to initialize.



ASL Recognition using Landmark-Based Pose Estimation in Real Time



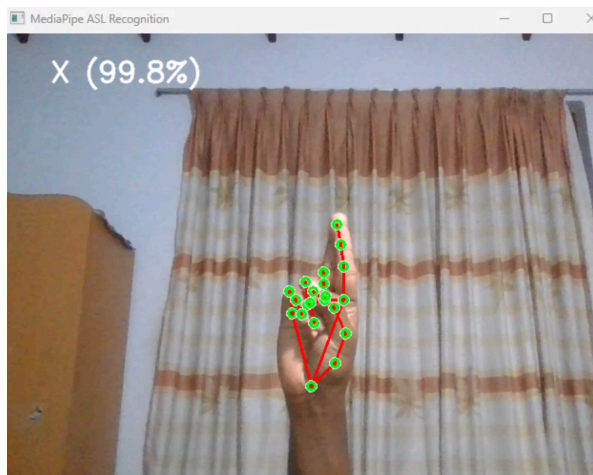
ASL Recognition using CNN with the 28x28x1 resolution dataset



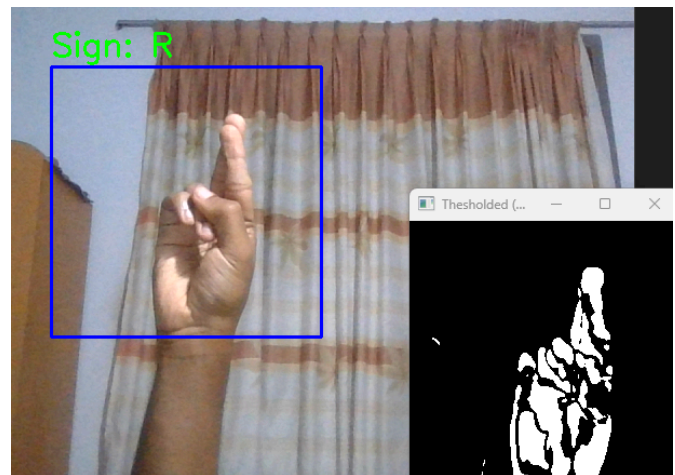
*ASL Recognition using MediaPipe + ANN with the 28x28x1 resolution dataset
Used 50 epochs*

2. Depth Perception and Occlusion MediaPipe estimates 3D coordinates from a 2D image, which causes it to struggle when one finger is hidden behind another.

- **What's missing:** In ASL, letters like 'R' (fingers crossed) or 'M/N' (thumb tucked) involve significant occlusion.
- **The Impact:** When a joint is hidden, MediaPipe must "guess" its location. A CNN doesn't need to find the joint; it recognizes the specific shape of the "hidden" finger, which is more reliable for these specific letters.



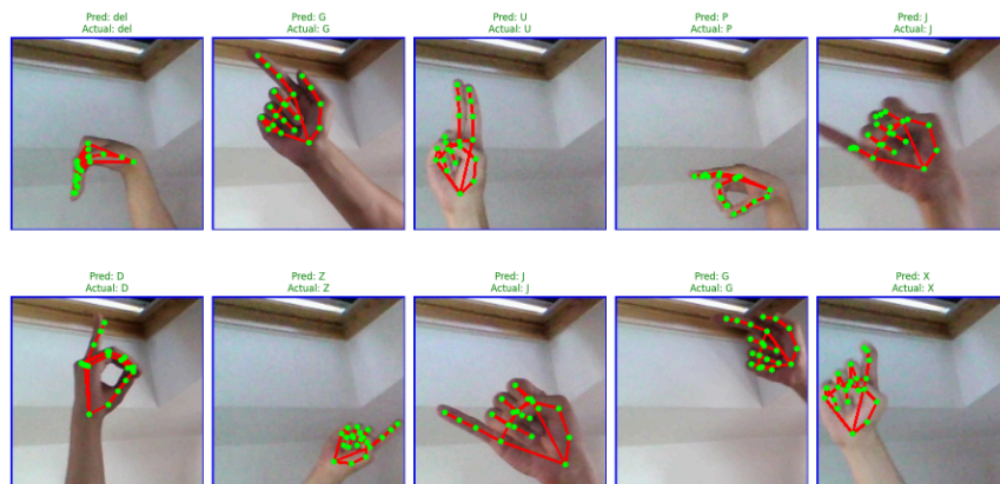
*Letter R detection
Using MediaPipe fails*



Using CNN for letter R detection

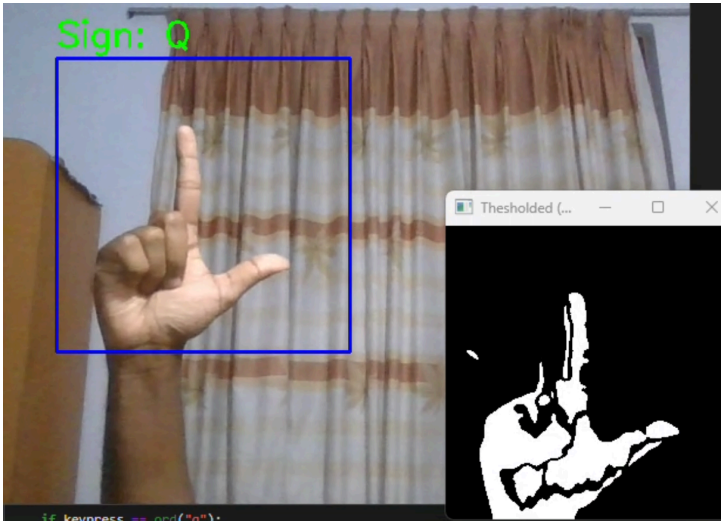
3. Lighting and Shadow Cues A CNN treats shadows as part of its "feature map", whereas MediaPipe attempts to ignore them to remain "robust."

- **What's missing:** Shadows provide significant information about the 3D volume and orientation of the hand and also due to stray light CNN might mislead.
- **The Impact:** By ignoring light, MediaPipe loses depth cues but identifies the sign correctly. CNN uses these shadows to understand the orientation of the palm, and also the body language like happy, sad....ect



ASL Recognition using MediaPipe + ANN with the 200x200x3 resolution dataset

Even though the original dataset provides 200x200x3 color images, landmark approach uses MediaPipe to abstract each frame into 21 structural points. These 21 landmarks are flattened into a 63-dimensional coordinate vector and fed into an Artificial Neural Network (ANN) to build the geometric model



CNN fails identifying 'L' Due to stray lightings

Feature	MediaPipe + ANN (Landmarks)	Pixel-Based CNN
Minimum Resolution	Struggles below 128x128	Efficient even at 28x28x1
Background Noise	Filters out the background entirely	Can be distracted by background clutter
Input Focus	Mathematical coordinates (Geometry)	Raw pixels (Light and Texture)
Real-Time Use	Best for high-res webcams	Even best for low-resolution static datasets

"Hit the Point"

According to this analysis we can see that even for **Sign MNIST (low quality images)**, using CNN is superior. However, for **Real-Time Webcam (high quality)**, MediaPipe is often preferred because it isn't distracted by your face or the background.

5. Use Case Scenarios: When to Use Which?

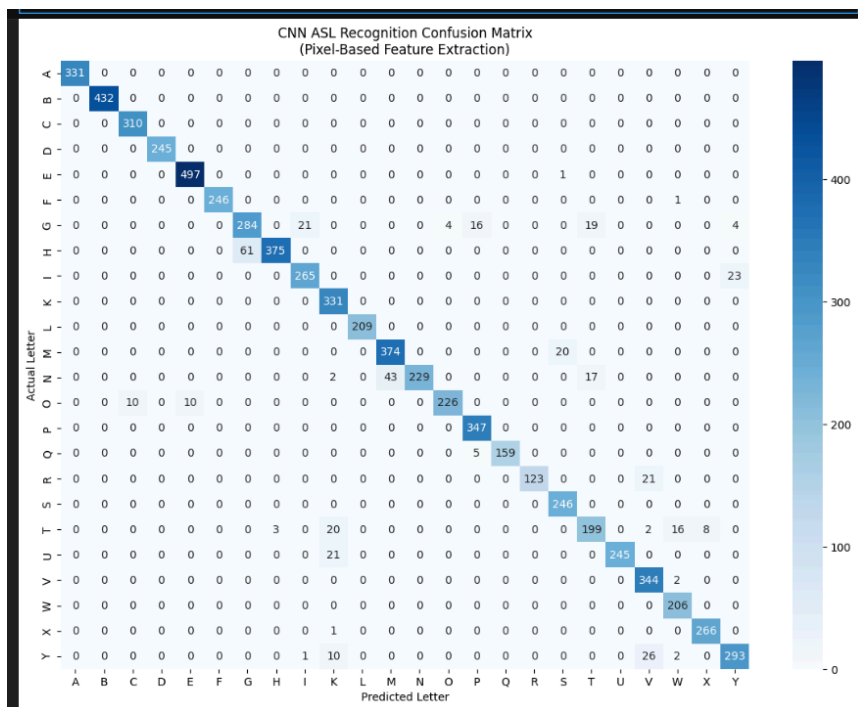
Based on my results, I have categorized into some cases as follows:

Scenario	Recommended Approach	Reason
Mobile Apps / Low-Power Hardware	Landmark Model	Extremely efficient; requires minimal RAM and battery power.
Real-Time Video Communication	Landmark Model	Low latency ensures the user sees results instantly as they sign.
High-End Research / Lab Settings	CNN Model	Capable of learning textures and skin folds that landmarks ignore. Because this model consider the 3 color channels in a pixel
Dynamic Backgrounds	Landmark Model	MediaPipe handles the segmentation, making it robust in diverse rooms.

6. Evaluation and Challenges

While training the CNN, I faced challenges with computational power, which initially limited my ability to reach high accuracy. This led me to appreciate the "elegance" of the Landmark approach. If a finger is hidden behind another, the landmark model can provide incorrect data due to visual ambiguity; however, a CNN may still recognize the gesture due to its higher spatial feature representation and ability to learn complex visual hierarchies from raw pixel data.

1. CNN ASL Recognition Model Using 28x28x1 Pixel Dataset

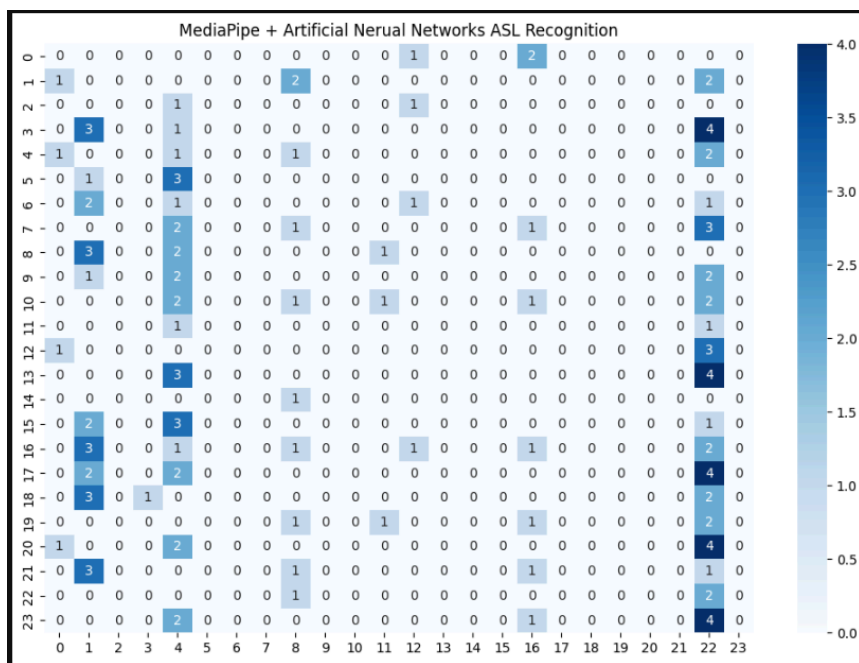


Classification Report:				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	331
1	1.00	1.00	1.00	432
2	0.97	1.00	0.98	310
3	1.00	1.00	1.00	245
4	0.98	1.00	0.99	498
5	1.00	1.00	1.00	247
6	0.82	0.82	0.82	348
7	0.99	0.86	0.92	436
8	0.92	0.92	0.92	288
10	0.86	1.00	0.92	331
11	1.00	1.00	1.00	209
12	0.90	0.95	0.92	394
13	1.00	0.79	0.88	291
14	0.98	0.92	0.95	246
15	0.94	1.00	0.97	347
16	1.00	0.97	0.98	164
17	1.00	0.85	0.92	144
18	0.92	1.00	0.96	246
19	0.85	0.80	0.82	248
20	1.00	0.92	0.96	266
21	0.88	0.99	0.93	346
22	0.91	1.00	0.95	206
23	0.97	1.00	0.98	267
24	0.92	0.88	0.90	332
accuracy			0.95	7172
macro avg	0.95	0.94	0.95	7172
weighted avg	0.95	0.95	0.95	7172

CNN ASL Recognition Confusion Matrix
(Pixel-Based Feature Extraction)

Classification Report

2. MediaPipe + ANN Using 28x28x1 Pixel Dataset

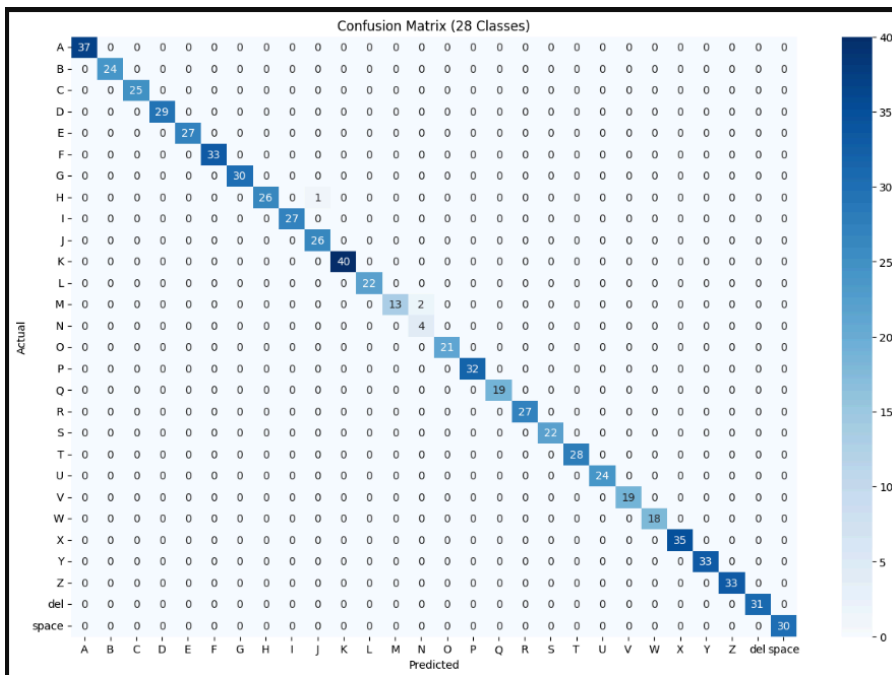


Classification Report:				
	precision	recall	f1-score	support
0	0.00	0.00	0.00	3
1	0.00	0.00	0.00	5
2	0.00	0.00	0.00	2
3	0.00	0.00	0.00	8
4	0.03	0.20	0.06	5
5	0.00	0.00	0.00	4
6	0.00	0.00	0.00	5
7	0.00	0.00	0.00	7
8	0.00	0.00	0.00	6
10	0.00	0.00	0.00	5
11	0.00	0.00	0.00	7
12	0.00	0.00	0.00	2
13	0.00	0.00	0.00	4
14	0.00	0.00	0.00	7
15	0.00	0.00	0.00	1
16	0.00	0.00	0.00	6
17	0.12	0.11	0.12	9
18	0.00	0.00	0.00	8
19	0.00	0.00	0.00	6
20	0.00	0.00	0.00	5
21	0.00	0.00	0.00	7
22	0.00	0.00	0.00	6
23	0.04	0.67	0.08	3
24	0.00	0.00	0.00	7
accuracy			0.03	128
macro avg	0.01	0.04	0.01	128
weighted avg	0.01	0.03	0.01	128

MediaPipe + Artificial Neural Networks ASL Recognition

Classification Report

3. MediaPipe + ANN Using 200x200x3 Pixel Dataset



Classification Report:				
	precision	recall	f1-score	support
A	1.00	1.00	1.00	37
B	1.00	1.00	1.00	24
C	1.00	1.00	1.00	25
D	1.00	1.00	1.00	29
E	1.00	1.00	1.00	27
F	1.00	1.00	1.00	33
G	1.00	1.00	1.00	30
H	1.00	0.96	0.98	27
I	1.00	1.00	1.00	27
J	0.96	1.00	0.98	26
K	1.00	1.00	1.00	40
L	1.00	1.00	1.00	22
M	1.00	0.87	0.93	15
N	0.67	1.00	0.80	4
O	1.00	1.00	1.00	21
P	1.00	1.00	1.00	32
Q	1.00	1.00	1.00	19
R	1.00	1.00	1.00	27
S	1.00	1.00	1.00	22
T	1.00	1.00	1.00	28
U	1.00	1.00	1.00	24
V	1.00	1.00	1.00	19
W	1.00	1.00	1.00	18
X	1.00	1.00	1.00	35
Y	1.00	1.00	1.00	33
Z	1.00	1.00	1.00	33
del	1.00	1.00	1.00	31
space	1.00	1.00	1.00	30
accuracy			1.00	738
macro avg	0.99	0.99	0.99	738
weighted avg	1.00	1.00	1.00	738

MediaPipe + Artificial Neural Networks ASL Recognition

Classification Report

7. Conclusion

Based on my comparative analysis it indicates that for an Accessibility Tool intended for daily use, the Landmark-based approach is superior. It offers the speed and reliability necessary for a nonverbal person to communicate without frustration. While the CNN remains a powerful "End-to-End" prototype for more advanced feature extraction, its hardware demands make it less practical for real-world edge deployment.

Through this analysis, I have gained a practical understanding of how computers "see" and interpret human gestures. By comparing two different systems, I explored how we can take complex image data and simplify it into mathematical skeletons to build faster, more efficient accessibility tools. I specifically focused on the trade-offs between speed and "visual intuition" understanding when a model needs to see every pixel and when it only needs to see the joints of a hand like structure

8. Technical References

- **Dataset:** ASL Alphabet Dataset (87,000 images, 29 classes)
<https://microsoft.github.io/data-for-society/dataset?d=MS-ASL-American-Sign-Language-Dataset>
- **MediaPipe:** <https://en.wikipedia.org/wiki/MediaPipe>
- **CNN:** https://en.wikipedia.org/wiki/Convolutional_neural_network
- **Tools:** Python, TensorFlow, Keras, OpenCV, MediaPipe.
<https://www.tensorflow.org/>
<https://opencv.org/>
<https://keras.io/>
- **Hardware Environment:** Local CPU & GPU.